

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15E.4

CHROMATOGRAPHY AND MASS SPECTROMETRY (GC-MS, HPLC-MS)

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: CARBOXYLIC ACID DERIVATIVES, SPECTROSCOPY AND CHROMATOGRAPHY
SUB-TOPIC:	15E.4: Chromatography and Mass Spectrometry (GC-MS, HPLC-MS)
TOTAL QUESTIONS:	1 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	3 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15E.4

Q1. 1. (a) Explain why gas chromatography combined with mass spectrometry (GC-MS) is superior to GC alone for identifying unknown substances in forensic analysis.

[3 Marks]**WCH14/01/Jan23/Q9(a)**

SECTION B: WORKED MARK SCHEME — 15E.4

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Explain why gas chromatography combined with mass spectrometry (GC-MS) is superior to GC alone for identifying unknown substances in forensic analysis. 1. GC separates complex mixtures into individual pure components. 2. MS provides molecular mass (M^+ peak) and characteristic fragmentation pattern for unambiguous structural identification. 3. GC alone only gives retention times, which may overlap for different compounds.	[3]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 15E.4**FAQ #1: Why is GC-MS widely used in drug testing and environmental monitoring?****Category: Analytical Applications****COMMON EXAMINER TRAP / MISTAKE:**

Giving general answers without mentioning separation vs identification.

MODEL ANSWER / EXAMINER EXPECTATION:

GC provides high separation efficiency for trace mixtures, while MS provides definitive structural confirmation by matching fragmentation patterns against spectral databases.